A CASE-STUDY BASED REPORT

ON

Personal Productivity Pattern Analyser

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Abstract

Personal Productivity Pattern Analyzer is a machine learning-based system designed to analyze an individual's daily activities and understand the factors that influence their productivity. Daily productivity can be affected by several factors such as study hours, sleep duration, phone and social media usage, entertainment time, exercise, and breaks. Manually identifying relationships between these factors and productivity can be difficult. The proposed system collects these activity parameters and analyzes them to identify meaningful productivity patterns.

The system estimates a user's productivity score and categorizes the result into different productivity levels. It also identifies common behavioral patterns among users and detects activity patterns that differ significantly from the usual trends. By combining these analyses, the system provides a broader understanding of how different daily habits are associated with productivity. The system is designed as an end-to-end application, connecting data processing, model development, evaluation, and prediction through a web-based interface.

The main objective of the project is to help users understand their daily routines and recognize habits that may support or reduce their productivity. The application also maintains productivity history, allowing users to compare their results over multiple check-ins and observe changes in their routine over time. This makes the system useful not only for predicting productivity but also for encouraging self-awareness and more consistent daily habits.

Introduction

Background

Productivity is influenced by a combination of daily habits and activities, including study duration, sleep, phone usage, social media, entertainment, exercise, and breaks. With students increasingly spending significant amounts of time on digital devices, understanding how these activities relate to productivity has become important. However, it can be difficult for individuals to identify meaningful patterns in their daily routines simply by observing their activities.

Traditional productivity tracking generally focuses on recording individual activities or maintaining simple task lists. Such approaches may show **what a person did**, but they do not necessarily help in understanding how different activities are associated with overall productivity. A data-driven approach can analyze multiple factors together and reveal relationships and patterns that may not be immediately noticeable.

The **Personal Productivity Pattern Analyzer** addresses this need by using historical activity data to analyze productivity and behavioral patterns. The system considers different daily activity parameters, estimates productivity levels, identifies common patterns, and detects unusual activity profiles. Through a web-based interface, users can record their daily activities, view productivity results, and maintain a history of their check-ins. This provides a structured way to understand personal routines and observe productivity changes over time.

Motivation for the Project

Students manage several activities throughout the day, including studying, sleeping, using mobile devices, taking breaks, exercising, and spending time on entertainment. These activities can vary considerably from one day to another and may be associated with differences in productivity. For example, a student may spend several hours studying but still experience low productivity because of insufficient sleep, excessive phone usage, or frequent interruptions. Similarly, a well-balanced routine involving adequate sleep, focused study time, exercise, and appropriate breaks may contribute to better productivity. However, identifying such relationships simply by observing daily routines can be difficult.

The increasing use of digital devices has made this problem more relevant. Students often spend considerable amounts of time on smartphones, social media, video platforms, and gaming in addition to their academic activities. Although these activities are not necessarily negative by themselves, their duration and combination with other daily habits can influence how a person's day is experienced and managed. Without systematically recording and analyzing these activities, it can be difficult to determine which patterns are associated with higher or lower productivity.

The motivation behind the **Personal Productivity Pattern Analyzer** is to provide a systematic and data-driven approach to understanding these daily activities. Instead of relying only on assumptions or general productivity advice, the system considers multiple activity factors together and analyzes their relationship with productivity. By using historical activity data, the system can identify patterns that may not be immediately visible through manual observation. This makes it possible to move from simply recording activities to gaining meaningful insights

Finally, the project provides an opportunity to demonstrate how data analysis and machine learning techniques can be applied to a practical everyday problem. The project combines data preparation, exploratory analysis, predictive modeling, behavioral pattern analysis, unusual activity detection, model evaluation, and application development into one system. Therefore, the motivation is not only to create a productivity tool for students but also to demonstrate the practical application of data-driven methods in understanding human activity patterns.

Overview of the work carried out

The **Personal Productivity Pattern Analyzer** was developed as an end-to-end system for analyzing the relationship between daily activities and productivity. The work began with understanding the available dataset and identifying the different parameters that could be used to represent an individual's daily routine. These parameters include study duration, sleep duration, phone usage, social media usage, YouTube usage, gaming, exercise, breaks, and other activity-related factors. The dataset was examined to understand its structure, distributions, and relationships with the productivity score.

The first stage of the work involved **data preparation and exploratory analysis**. The available data was inspected for missing values, duplicate records, inconsistent values, and other issues that could affect subsequent analysis. The relevant features were selected and prepared for further processing. Exploratory analysis was then performed using statistical summaries, distributions, correlation analysis, and visualizations. These analyses helped identify important relationships between activity factors and productivity and provided an initial understanding of the dataset before model development.

The next stage focused on **productivity prediction**. Since productivity is represented by a numerical score, prediction was treated primarily as a regression problem. Multiple regression approaches were developed and evaluated to determine how effectively they could estimate productivity from the available activity information. Appropriate evaluation measures were used to compare the models and identify a suitable approach for the application. The predicted numerical score was then mapped into understandable productivity levels such as Low, Medium, and High.

The developed analytical components were then integrated into a **FastAPI-based backend**. The backend receives activity information from the user interface, processes the input using the trained model, and returns the estimated productivity score and corresponding productivity level. This creates a connection between the data analysis and the actual application, allowing the trained system to be used interactively rather than only through standalone scripts.

Finally, a **web-based user interface** was developed to make the system easier to use. Users can record their daily activities through a check-in interface, obtain their productivity result, and maintain a history of previous check-ins. Additional sections such as goals and insights are included to support continued interaction with the application. Overall, the work combines data preparation, exploratory analysis, predictive modeling, behavioral analysis, unusual activity detection, evaluation, backend integration, and frontend development into a single productivity analysis system.

Scope and Responsible-Use Statement

The scope of the **Personal Productivity Pattern Analyzer** is to provide a data-driven system for understanding how recorded daily activities are associated with productivity. The system focuses on activity-related information such as study hours, sleep, phone usage, social media usage, entertainment activities, exercise, and breaks. By analyzing these factors together, the application provides an estimated productivity score and a corresponding productivity level based on the patterns learned from the available dataset.

The project also extends beyond simple productivity prediction by incorporating **behavioral pattern analysis and unusual activity detection**. Behavioral analysis allows the system to identify groups of activity profiles that share similar characteristics, while unusual activity detection helps identify profiles that differ considerably from the common patterns in the data. These components provide additional information that can help users understand their routines from different perspectives.

Another part of the project's scope is the development of an accessible application rather than limiting the work to offline data analysis. The trained models are integrated into a backend service through FastAPI, while the frontend provides an interface for entering daily activity information and viewing results. The application also supports recording previous check-ins so that users can observe their productivity results over time. This creates a practical connection between the analytical components and everyday user interaction.

Responsible use is an important consideration because productivity is influenced by many personal, environmental, and situational factors that may not be represented in the dataset. The system should therefore not be used to judge a person's ability, discipline, intelligence, academic performance, or character. Similarly, its results should not be used for employment decisions, academic grading, medical diagnosis, or other high-stakes decisions. A productivity score should be treated as an **estimate rather than an objective measurement of a person's overall performance**.

The reliability of the system also depends on the quality and representativeness of the data used during development. Differences between the dataset and a real user's circumstances may affect the accuracy of the predictions or patterns identified by the system. Users should therefore consider the results as supportive information rather than absolute conclusions. The intended outcome is to encourage awareness of daily routines and provide useful data-driven observations while maintaining appropriate limitations on how the system's results are interpreted and used.

Problem Statement

The increasing use of digital devices, changing study routines, irregular sleep patterns, and differences in daily habits make it difficult for students to understand what factors are associated with their productivity. Although students may record individual activities such as study time or screen time, simply collecting these values does not provide a clear understanding of how different activities relate to overall productivity. There is therefore a need for a system that can analyze multiple daily activity factors together and provide meaningful productivity-related insights.

The **Personal Productivity Pattern Analyzer** addresses this problem by using historical activity data to estimate productivity and identify patterns in daily routines. The system considers factors such as study hours, sleep, phone usage, social media, entertainment, exercise, and breaks. In addition to estimating a productivity score, the system analyzes behavioral patterns and identifies activity profiles that differ from common patterns.

The central problem is therefore to develop a practical and understandable system that can transform daily activity information into productivity-related insights while recognizing the limitations of data-driven predictions. The system should provide useful results without treating the predicted productivity score as an absolute measure of an individual's capabilities or performance.

Case-Study Scenario

Consider a student who follows a different routine on different days. On one day, the student may study for several hours but also spend considerable time on the phone and get insufficient sleep. On another day, the student may study for fewer hours but sleep adequately, take regular breaks, exercise, and spend less time on distracting activities. The student may notice that the quality of the day feels different but may not be able to identify which combination of activities is associated with the difference.

In the proposed case study, the student records relevant daily activities through the **Personal Productivity Pattern Analyzer**. Information such as study duration, sleep duration, phone usage, social media usage, YouTube or gaming time, exercise, and breaks is provided through a daily check-in. The system processes this information and produces an estimated productivity score along with a productivity level.

As more check-ins are recorded, the system can provide a broader view of the student's routine. Previous results can be compared to identify changes over time, while behavioral pattern analysis can reveal similarities between different activity profiles. Unusual activity detection can also highlight days whose activity combinations differ considerably from common patterns.

This case study demonstrates how a student's routine can be analyzed systematically rather than relying entirely on personal assumptions. The purpose is not to determine whether a student is inherently productive or unproductive, but to provide information that can support greater awareness of their daily habits.

Central Case-Study Question

The central question guiding this study is:

> **How can daily activity patterns be analyzed to estimate productivity and identify behavioral patterns that may help individuals better understand their routines?**

This question focuses on the relationship between measurable daily activities and productivity. Rather than examining a single factor in isolation, the study considers multiple aspects of a person's routine together. This allows the system to investigate whether combinations of activities can provide useful information about productivity levels.

The question also extends beyond simple prediction. The study examines whether activity profiles can be grouped into recognizable behavioral patterns and whether significantly different activity profiles can be identified as unusual. In this way, the project considers productivity from multiple perspectives: estimation of the productivity score, identification of routine patterns, and detection of activity profiles that differ from common behavior.

The case study therefore evaluates whether a data-driven system can provide useful and understandable insights from everyday activity information while maintaining appropriate limitations on interpretation.

Formal Problem Definition

Let the daily activity information of an individual be represented by a feature vector:

$$X = \{x_1, x_2, ..., x_n\}$$

where the features represent measurable aspects of the individual's daily routine, such as study hours, sleep hours, phone usage, social media usage, entertainment time, exercise, and breaks.

The primary objective is to learn a function:

$$f(X) \rightarrow Y$$

where X represents the recorded activity information and Y represents the estimated productivity score.

The system then maps the estimated score into a productivity category using predefined ranges:

$$Y < 40 \rightarrow Low$$

$$40 \leq Y < 70 \rightarrow Medium$$

$$Y \geq 70 \rightarrow High$$

In addition to productivity estimation, the problem includes identifying groups of similar activity profiles. Given a collection of activity vectors $X_1, X_2, ..., X_m$, the system analyzes their similarities to discover common behavioral patterns. The system also identifies activity profiles that significantly differ from the general distribution of the available data.

Thus, the formal problem consists of three related tasks: **estimating productivity from daily activity information, identifying common behavioral patterns, and detecting unusual activity profiles**. The final system integrates these analytical tasks into an application that allows users to record and review their activity information.

Objectives that Guide the Study

The primary objective of the project is to develop a system capable of analyzing daily activity information and estimating an individual's productivity based on patterns learned from historical data.

The specific objectives of the study are:

1. **To prepare and analyze activity-based productivity data** by examining its structure, quality, distributions, and relationships between different variables.
2. **To identify important relationships between daily activities and productivity**, including factors such as study duration, sleep, phone usage, entertainment, exercise, and breaks.
3. **To develop and evaluate productivity prediction models** that can estimate a numerical productivity score from the available activity information.
4. **To categorize productivity results** into understandable levels such as Low, Medium, and High for easier interpretation by users.
5. **To identify common behavioral patterns** by analyzing similarities between different daily activity profiles.
6. **To detect unusual activity profiles** that differ noticeably from common patterns in the available dataset.
7. **To integrate the analytical components into a web-based application** so that users can enter their daily activity information and receive results interactively.
8. **To maintain productivity history** so users can observe their results across multiple check-ins and identify changes in their routines over time.
9. **To provide understandable and responsible insights** without presenting productivity predictions as definitive judgments about an individual.

Assumptions and Limitations

The project assumes that the activity information provided by users is reasonably accurate and represents their actual daily routine. It also assumes that the patterns present in the available dataset contain sufficient information to establish useful relationships between activity factors and productivity. The system assumes that the selected activity variables provide meaningful information for productivity analysis, although productivity can also be influenced by factors that are not included in the dataset.

One important assumption is that the productivity score used by the system is a suitable representation of the target being analyzed. The productivity levels used by the application are derived from predefined score ranges and are intended to provide an understandable interpretation of the numerical prediction. These categories should therefore be viewed as application-level interpretations rather than universally defined productivity standards.

The project has several limitations. The available dataset may not represent every type of student, lifestyle, academic environment, or personal circumstance. As a result, the patterns learned by the system may not generalize equally well to every individual. The system also relies on user-provided information, meaning inaccurate, incomplete, or inconsistent inputs can affect the resulting prediction.

Another limitation is that the system identifies **associations rather than causal relationships**. For example, if higher study hours are associated with higher productivity in the dataset, this does not prove that increasing study hours alone will cause productivity to increase for every individual. Other factors may influence the observed relationship.

Finally, the system should not be treated as a medical, psychological, academic, or employment assessment tool. Productivity is a complex concept that cannot be completely represented by a numerical score. The predictions, behavioral patterns, and unusual-activity results should therefore be interpreted as data-driven observations that support personal reflection rather than definitive conclusions about a person's abilities, health, discipline, or overall performance.

Dataset and Exploratory Data Analysis

About the Dataset

The dataset used in the **Personal Productivity Pattern Analyzer** contains individual-level daily activity and productivity information. The cleaned dataset focuses on measurable aspects of a person's daily routine and uses **productivity score** as the target variable. The activity variables represent time allocation, rest, digital-device usage, physical activity, and break patterns. The cleaned training data contains the following attributes: `study_hours_per_day`, `sleep_hours`, `phone_usage_hours`, `social_media_hours`, `youtube_hours`, `gaming_hours`, `exercise_minutes`, `breaks_per_day`, and `productivity_score`. cleaned_train

The selected variables represent different dimensions of a daily routine. Study hours represent the amount of time spent studying, while sleep hours represent the duration of rest. Phone usage, social media, YouTube, and gaming represent different forms of digital or entertainment activity. Exercise is recorded in minutes, and breaks represent the number of breaks taken during the day. The `productivity_score` is the target outcome that the system attempts to estimate from these activity-related variables. cleaned_train

The dataset is suitable for studying productivity because it combines multiple lifestyle factors rather than focusing on a single activity. This allows the project to investigate how different combinations of daily activities are associated with productivity. The data contains numerical measurements in different units, including hours, minutes, counts, and productivity-score values. The available dataset does not explicitly provide personal identities, dates, departments, or other identifying information. cleaned_train

For the project, a cleaned version of the dataset was used for analysis and subsequent model development. The same activity structure is maintained between the training and testing data so that the trained system can later process new activity information in the same format.

Purpose of EDA

Exploratory Data Analysis (EDA) was performed to develop an initial understanding of the dataset before applying predictive and pattern-analysis techniques. The purpose of EDA was not only to examine individual variables but also to investigate how different daily activities are associated with the productivity score.

The first purpose of EDA was to understand the **distribution of the target variable**. Examining the productivity-score distribution helps determine whether the scores are concentrated around a particular range or whether the dataset contains a wide range of productivity levels. This provides useful context for later productivity prediction and categorization.

The second purpose was to investigate the relationship between individual activity factors and productivity. Variables such as study hours, sleep hours, phone usage, social media usage, YouTube usage, gaming, exercise, and breaks were examined against the productivity score. This helps identify which variables show stronger visible relationships with the target and which variables appear to have weaker relationships.

Another purpose of EDA was to identify potential data-quality issues before model development. The data was examined for missing or inconsistent observations and prepared into a cleaner form for subsequent processing. Visualization was also used to make relationships easier to interpret than through numerical summaries alone.

Finally, EDA provided a foundation for the later stages of the project. The observations obtained from the exploratory analysis helped guide feature selection, model development, evaluation, behavioral analysis, and interpretation of the final system.

Preparation Steps Actually Performed

The dataset was first examined to understand its available columns and identify the variables relevant to the productivity-analysis problem. The activity-related variables were retained as the main inputs, while `productivity_score` was treated as the target variable. The cleaned dataset therefore focuses specifically on the daily activity factors used by the application. cleaned_train

The data was then checked and prepared before exploratory analysis. The cleaning process produced dedicated cleaned training and testing files, namely `cleaned_train.csv` and `cleaned_test.csv`, which are used within the project. The resulting training data contains the activity variables and the productivity target required for analysis and model development.

The input variables were also kept in appropriate numerical form so that they could be used for statistical analysis, visualization, and machine-learning algorithms. Variables measured in hours, such as study, sleep, phone usage, social media, YouTube, and gaming, were treated as numerical values. Exercise was represented in minutes, breaks were represented as counts, and productivity was represented as a numerical score. cleaned_train

After preparation, the cleaned dataset was used for visualization and relationship analysis. Separate plots were generated for important activity variables against productivity, together with a productivity distribution and correlation analysis. These preparation steps ensured that the data used during EDA was consistent with the data expected by the subsequent modeling pipeline.

EDA Exploration – Implementation

The EDA implementation was carried out using Python-based data-analysis and visualization tools. The dataset was loaded and examined using tabular data-processing operations, after which graphical visualizations were generated to investigate the relationships between activity variables and productivity.

A **productivity distribution plot** was created to understand how the target productivity scores are distributed across the dataset. This provides an overall view of the target variable and helps identify whether the dataset contains a broad range of productivity outcomes.

Scatter-based visualizations were also generated for individual activity variables. These included:

- Study hours vs. productivity
- Sleep hours vs. productivity
- Phone usage vs. productivity
- Social media usage vs. productivity
- YouTube usage vs. productivity
- Gaming vs. productivity
- Exercise minutes vs. productivity
- Breaks per day vs. productivity

These visualizations make it possible to observe whether changes in an activity variable are accompanied by visible changes in productivity. They also help identify potential nonlinear relationships or widely scattered observations that may be difficult to recognize from summary statistics alone.

A **correlation analysis** was additionally performed to quantify the linear association between numerical variables and the productivity score. A correlation heatmap was generated to provide a visual representation of relationships between the activity variables and the target. The project also generated a target-correlation output to support further interpretation of the exploratory results.

The EDA outputs were saved as graphical files in the project's output directory. These visualizations were subsequently used to understand the dataset and support decisions made during model development and evaluation.

Summary of EDA Findings

The exploratory analysis indicates that productivity is related to multiple aspects of the recorded daily routine rather than being determined by a single activity. Among the variables examined, **study hours show a strong positive relationship with productivity**, making study duration one of the most prominent activity-related factors in the dataset.

Sleep duration also shows a positive association with productivity, although the relationship is weaker than that observed for study hours. This suggests that rest is relevant when examining productivity patterns, while also demonstrating that productivity cannot be explained by study time alone.

In contrast, **phone usage** shows a noticeable negative association with productivity in the exploratory analysis. This indicates that higher phone usage tends to occur alongside lower productivity scores in the available data. Social media usage, YouTube usage, and gaming show weaker relationships individually, suggesting that their influence may be more difficult to interpret when considered separately.

The analysis also shows that **exercise and breaks** have more complex relationships with productivity. Their effect cannot be adequately described by simply assuming that more or fewer breaks or exercise minutes always correspond to higher productivity. This supports the decision to consider multiple variables together rather than relying on simple individual-factor rules.

Overall, the EDA demonstrates that productivity is associated with a combination of study, rest, digital usage, and daily activity patterns. The findings provided a basis for proceeding to predictive modeling and behavioral analysis. At the same time, the observed relationships represent **associations within the dataset and should not be interpreted as proof that one activity directly causes a change in productivity**. This distinction is important when interpreting the results of the complete system.

Algorithm Selection and Implementation

1. Algorithm Selection

The **Personal Productivity Pattern Analyzer** uses a combination of supervised and unsupervised learning techniques to analyze student activity patterns. The algorithms were selected based on the nature of the dataset, the project objectives, and the need to obtain understandable productivity results.

For the main productivity prediction task, **Gradient Boosting Regression** was selected. The target variable, `productivity_score`, is a continuous value, making regression appropriate. Gradient Boosting Regression combines multiple decision trees sequentially, where each new tree attempts to reduce the errors made by the previous trees. It is suitable for this project because productivity can depend on several interacting activity factors such as study time, sleep, phone usage, entertainment time, exercise, and breaks.

For discovering groups of students with similar activity patterns, **K-Means Clustering** was implemented. It groups observations according to the similarity of their daily activity characteristics. This allows the system to identify different routine patterns without requiring predefined labels.

For detecting unusual activity profiles, **Isolation Forest** was implemented. It identifies observations that differ significantly from the majority of the dataset. This supports the project's unusual-activity analysis by highlighting activity combinations that are less common among the recorded student routines.

Selected Algorithms

Purpose	Algorithm	Reason for Selection
Productivity prediction	Gradient Boosting Regression	Suitable for continuous productivity scores and nonlinear relationships
Behavior pattern discovery	K-Means Clustering	Groups students with similar activity patterns
Unusual activity detection	Isolation Forest	Detects activity profiles that differ from common patterns

2. Productivity Prediction – Gradient Boosting Regression

The main prediction component uses **Gradient Boosting Regression** to estimate a productivity score between 0 and 100.

The model uses the following daily activity features:

- Study hours per day
- Sleep hours
- Phone usage hours
- Social media hours
- YouTube hours
- Gaming hours
- Exercise minutes
- Breaks per day

The model is trained using the cleaned training dataset. After training, the model is saved using `joblib` so that it can be loaded by the FastAPI application without retraining every time the application starts.

```
import pandas as pd

import joblib



from sklearn.model_selection import train_test_split

from sklearn.ensemble import GradientBoostingRegressor



# Load cleaned dataset

df = pd.read_csv("data/raw/cleaned_train.csv")



# Selected input features

FEATURES = [

    "study_hours_per_day",

    "sleep_hours",

    "phone_usage_hours",
```

```
    "social_media_hours",
    "youtube_hours",
    "gaming_hours",
    "exercise_minutes",
    "breaks_per_day"
]

TARGET = "productivity_score"

# Separate input and target
X = df[FEATURES]
y = df[TARGET]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

# Create Gradient Boosting model
model = GradientBoostingRegressor(
    random_state=42
)
```

```
# Train model
model.fit(X_train, y_train)


# Save trained model
joblib.dump(
    model,
    "models/gradient_boosting_regression.joblib"
)


print("Gradient Boosting Regression model trained successfully.")
```

The dataset is divided into training and testing portions so that the model can be evaluated on data that was not used during training. The trained model is then stored in the `models` directory.

3. Productivity Level Generation

Although the model produces a continuous productivity score, the application also converts the score into three understandable productivity levels.

The thresholds used in the application are:

- **Below 40:** Low
- **40 to below 70:** Medium
- **70 and above:** High

Implementation

```
if score < 40:
    level = "Low"
elif score < 70:
    level = "Medium"
else:
    level = "High"
```

This allows the application to present the prediction in a simple form such as:

Productivity Score: 72.45

Productivity Level: High

4. Behavior Pattern Analysis – K-Means Clustering

K-Means clustering is used to identify groups of students with similar daily routines.

The clustering process considers activity-related variables such as study, sleep, phone usage, entertainment, exercise, and breaks. Since these variables have different numerical ranges, scaling is performed before clustering.

Implementation

```
import pandas as pd

import joblib



from sklearn.preprocessing import StandardScaler

from sklearn.ensemble import IsolationForest



df = pd.read_csv("data/raw/cleaned_train.csv")



FEATURES = [

    "study_hours_per_day",

    "sleep_hours",

    "phone_usage_hours",

    "social_media_hours",

    "youtube_hours",

    "gaming_hours",

    "exercise_minutes",

    "breaks_per_day"

]



X = df[FEATURES]
```

```
# Scale the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Create Isolation Forest
model = IsolationForest(
    contamination=0.05,
    random_state=42
)

# Detect normal and unusual observations
anomaly = model.fit_predict(X_scaled)

result = df.copy()
result["anomaly"] = anomaly

# Convert numerical output into readable labels
result["status"] = result["anomaly"].map({
    1: "Normal",
    -1: "Unusual"
})

result.to_csv(
    "outputs/anomaly_results.csv",
    index=False
```

```
)


# Save models
joblib.dump(
    scaler,
    "models/anomaly_scaler.joblib"
)


joblib.dump(
    model,
    "models/isolation_forest.joblib"
)


print("Anomaly detection completed.")
print(result["status"].value_counts())
```

The resulting clusters can be examined to understand different routine patterns. For example, one group may contain students with relatively high study time, while another may contain students with higher entertainment or phone usage.

5. Unusual Activity Detection – Isolation Forest

Isolation Forest is used to identify activity profiles that are significantly different from the majority of the dataset.

Unlike the productivity prediction model, this component does not require a target variable. It examines the activity characteristics and determines whether an observation appears normal or unusual.

Implementation

```
import pandas as pd

import joblib


from sklearn.preprocessing import StandardScaler

from sklearn.ensemble import IsolationForest


df = pd.read_csv("data/raw/cleaned_train.csv")


FEATURES = [

    "study_hours_per_day",

    "sleep_hours",

    "phone_usage_hours",

    "social_media_hours",

    "youtube_hours",

    "gaming_hours",

    "exercise_minutes",

    "breaks_per_day"

]


X = df[FEATURES]


# Scale the features

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

```
# Create Isolation Forest
model = IsolationForest(
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    random_state=42
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# Detect normal and unusual observations
anomaly = model.fit_predict(X_scaled)

result = df.copy()
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result["status"] = result["anomaly"].map({
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    scaler,
    "models/anomaly_scaler.joblib"
)
```

```
joblib.dump(
    model,
    "models/isolation_forest.joblib"
)


print("Anomaly detection completed.")
print(result["status"].value_counts())
```

In the implemented dataset, the anomaly detection process produced **15,200 Normal observations and 800 Unusual observations**. These results are used for the project's unusual-activity analysis.

6. Model Integration with the Application

The trained Gradient Boosting model is connected to the application through **FastAPI**. When a user submits a daily check-in, the entered values are sent to the `/predict` endpoint.

FastAPI Implementation

```
@app.post("/predict")
def predict(req: PredictionRequest):

    if not MODEL.exists():
        return {
            "error": "Model not found. Run python src/train.py first."
        }

    X = pd.DataFrame([req.model_dump()])

    model = joblib.load(MODEL)

    score = float(model.predict(X)[0])

    score = max(0.0, min(100.0, score))

    if score < 40:
        level = "Low"
    elif score < 70:
        level = "Medium"
    else:
        level = "High"
```

```
    return {
        "productivity_score": round(score, 2),
        "productivity_level": level
    }
```

The overall implementation therefore follows the flow:

Daily Activity Input → Gradient Boosting Regression → Productivity Score → Productivity Level

while the additional analysis components follow:

Activity Data → K-Means → Behavior Patterns

and

Activity Data → Isolation Forest → Normal / Unusual Activity

This combination allows the project to provide not only a productivity score, but also broader analysis of student activity patterns.